

3D Point Generation and Projection Project Documentation:

Project Overview

This project generates 1,000 random 3D points using NumPy, stores them in JSON format, and visualizes them in a 3D plot. It also explains how 3D points can be rotated and projected onto a 2D plane using orthographic and perspective projection.

Workflow of the Project

1. Generate random X, Y, Z coordinates.
2. Combine coordinates into 3D points.
3. Save points to a JSON file.
4. Apply optional Yaw, Pitch, and Roll rotations.
5. Convert rotation angles from degrees to radians.
6. Project rotated 3D points to 2D using orthographic or perspective projection.
7. Display or use projected points.

Random Number Generation for 3D Points

NumPy's `np.random.uniform()` is used.

X: -200 to 200

Y: -400 to 400

Z: -100 to 100

Total points: 1000.

Rotation Formulas

Yaw (Y-axis): $x' = x \cos\theta + z \sin\theta$, $z' = -x \sin\theta + z \cos\theta$.

Pitch (X-axis): $y' = y \cos\theta - z \sin\theta$, $z' = y \sin\theta + z \cos\theta$.

Roll (Z-axis): $x' = x \cos\theta - y \sin\theta$, $y' = x \sin\theta + y \cos\theta$.

Degree to Radian Conversion

Formula: $\text{radians} = \text{degrees} \times \pi / 180$.

Purpose: Python trigonometric functions use radians instead of degrees.

Orthographic Projection

Projects 3D points without perspective distortion.

Formula: $X=x$, $Y=y$. Object size remains constant regardless of distance.

Perspective Projection

Objects farther away appear smaller.

Formula: $X = (f \cdot x) / (z + f)$, $Y = (f \cdot y) / (z + f)$, where f is focal length.

Overall Implementation

The program generates random coordinates, stores them as dictionaries, writes them to `dat.json`, and visualizes them. Rotation formulas can be applied before projecting the points to a 2D plane using orthographic or perspective projection.